Advancing the 3Rs (Reduce, Reuse, Recycle) through the Refurbishment of Medical Aids

By PILLAR 2025 (Alyaa, Amira & Danial)

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1. Introduction

The concept of advancing the 3Rs (Reduce, Reuse, Recycle) through the refurbishment of medical aids was conceptualised during the rotation at *Pusat Rehabilitasi PERKESO Tun Abdul Razak*. During the exposure to the supply process of medical aids, such as wheelchairs, to *Orang Berinsurans (OBs)*, it was observed that no structured mechanism currently exists for retrieving equipment once it is no longer in use. For instance, in cases where an OB requires a wheelchair temporarily due to a fractured leg and subsequently recovers, the equipment often remains unused without a formal process for return or redistribution, potentially contributing to avoidable waste generation.

** TALK ABOUT FUND SUSTAINABILITY – ILO SESSION

The absence of a retrieval framework presents significant implications for PERKESO's Environmental, Social and Governance (ESG) footprint, particularly in relation to medical waste generation and inefficient resource utilisation.

In addition, with the expansion of protection coverage of *Skim Kemalangan Bukan Bencana Kerja (SKBBK)* which will be providing 24-hour coverage including for foreign workers, the projected volume of claims and rehabilitation cases is expected to increase. This expanded coverage will inevitably increase the demand for medical aids.

As coverage expands and the insured population grows, reliance on a traditional one-time supply approach will place increasing pressure on asset management systems and storage capacity. A more sustainable and circular framework is therefore required to ensure long-term operational resilience.

International benchmarking indicates that circular rehabilitation models are both feasible and sustainable. A notable example is the Accident Compensation Corporation (ACC) of New Zealand which operates a national no-fault injury insurance scheme. Under the ACC model:

- Rehabilitation equipment remains part of the ACC-funded system.
- Equipment is not permanently owned by patients in most cases.
- Items are retrieved, refurbished, and redistributed.

This approach represents a circular rehabilitation equipment model rather than a one-time supply model.

In contrast, PERKESO's current framework operates largely as a linear supply mechanism, where medical aids are issued but not systematically tracked throughout their lifecycle. The absence of retrieval and monitoring structures results in what may effectively be described as a "supply-and-forget" model. Transitioning away from this linear approach towards a lifecycle-managed circular system is therefore both an operational and governance imperative.

2. Background: R-Pemulihan Project from Segamat Local Office

During the rotation in the Audit Team, it was highlighted by *pegawai tadbir*, Pn Nurulhuda Binti Yaakub, that PERKESO Segamat had previously initiated a similar project known as *R-Pemulihan* in 2017. Consultation was subsequently conducted with PERKESO Segamat *Pegawai Eksekutif Kanan*, En Mohd Syahmi Bin Samsuddin, who pioneered *R-Pemulihan*. It was confirmed that the project did not achieve full implementation due to structural and governance constraints.

Figure 1: Virtual Session with Mr. Mohd Syahmi from PP Segamat

2.1 Identified Constraints

1. Absence of Official Directives

A major barrier was identified in the absence of formal circulars *(pekeliling)* or gazetted rules explicitly authorising the retrieval of medical aids after issuance. Without written procedural authority, implementation resistance was observed across regional offices.

2. Audit Concerns

Internal auditors raised concerns regarding the storage of medical aids within PERKESO premises when such items were no longer recorded as PERKESO assets. The absence of clear classification between “issued assets” and “recoverable assets” resulted in accounting ambiguity.

3. Storage and Logistics Limitations

Temporary storage at Segamat Hospital was no longer available. Transportation assistance had previously been provided by Segamat Local Office. However, resistance was observed due to concerns relating to manpower allocation and logistical costs.

4. Managerial Resistance

Although conceptual support was shown by senior management, reluctance was observed among middle management and veteran officers in the absence of formalised Standard Operating Procedures (SOPs) and written directives.

3. Sessions with Putrajaya Local Office

Pn Norliza Binti Noordin, *Pengurus Pejabat PERKESO Putrajaya*, has previously worked closely with the *R-Pemulihan* initiative. She was therefore appointed as the supervisor of the present "Advancement of 3Rs and Refurbishment of Medical Aids" project.

Data has been collected from *BPPP (Bahagian Pencegahan, Perubatan, Pemulihan)* in order to project the potential reduction in plastic wastage and the amount of time that may be saved through refurbishment practices. The projections will be presented in the subsequent sections of this report.
DATA COLLECTED FROM REHAB TOO

** put En Samah + En Din names for providing data and pics

With the assistance of case managers (CM) and *Skuad Sinergi* from the Putrajaya Local Office, an improved framework has been developed which will serve as the structural backbone of the entire project as seen below:

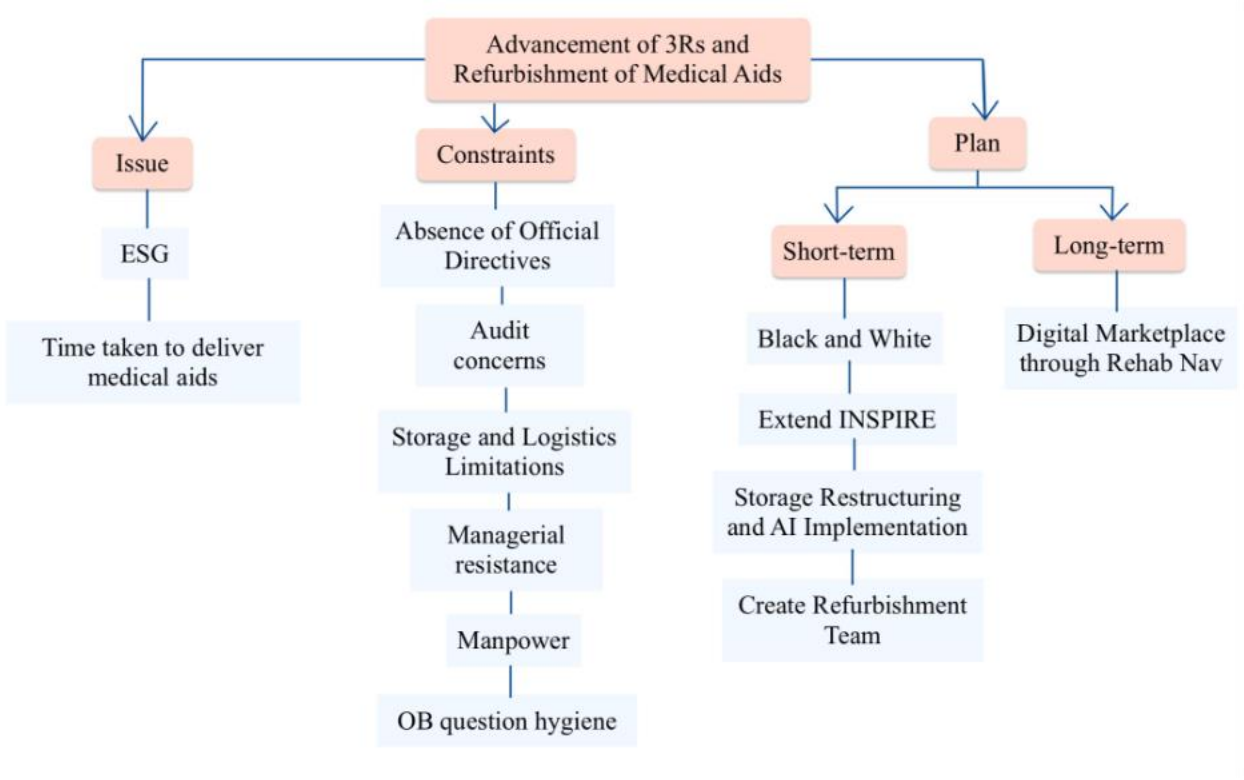

Figure 2: Capstone Project Framework

The officers involved are Pn Meilin Lampisung *(Pegawai Pengurusan Hilang Upaya)*, Pn Noor Syazwani Binti. Azmi *(Penolong Pegawai Pengurusan Hilang Upaya)*, En Ahmad Sapawie Bin Rokman *(Pegawai Eksekutif)*, Pn Mastura Bt Omar *(Pegawai Eksekutif Kanan)*, En Marwan Bin. Ibrahim *(Pegawai Eksekutif),* and En Mohd Ashreeq Amierullah Bin. Mohd Firdaus *(Pegawai Eksekutif)*.

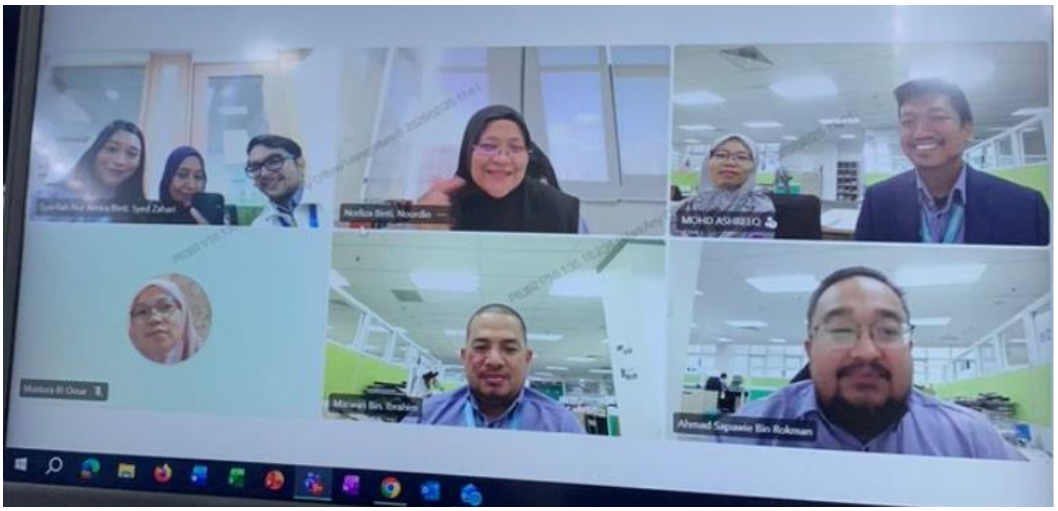

Figure 3: First Virtual Meeting with PP Putrajaya

Figure 4: Second In-person Meeting at Putrajaya Local Office

We have collected data from *BPPP (Bahagian Pecegahan, Perubatan, Pemulihan)* to project how much we can plastic wastage we can prevent and the amount of time we can save by refurbishing.

DATA COLLECTED FROM REHAB TOO

The projections can be seen as below:

PROJECTION

4. Proposed Solutions

Although the initiative is not entirely new, further research has been conducted and additional planning has been undertaken to ensure that this project is feasible for implementation under current conditions and remains sustainable, strong and relevant in the future.

4.1 Short-term Plans (Month 0–24)

4.1.1 Phase 1 (Month 0–3): Black and White

Currently, medical aids are issued under **Section 57** of **Act 4**:

Section 57(1): an insured person suffering from or claiming to suffer from invalidity or permanent disability may be provided by the organisation, free of charge for physical or vocational rehabilitation.

Section 57(2): facilities under subsection (1) shall be of such nature and shall be provided to such insured persons and on such conditions as may be specified by the regulations.

Section 57(3): an insured person suffering from or claiming to suffer from invalidity or permanent disablement may, if his condition so requires, be provided free of charge with prosthetic, orthotic or other appropriate appliances as may be determined by the organisation and such appliances may be renewed, when necessary, free of charge. Double check

Observed Regulatory Gaps
From the above provisions, it is noted that there is:

- No provision stating that equipment must be new.
- No regulatory clause expressly permitting retrieval of unused equipment.
- No governance structure governing the implementation of refurbished medical aids.
- No formal asset classification distinguishing permanently issued equipment from recoverable equipment.

- No written authority enabling nationwide enforcement and compliance.

This creates legal flexibility under Section 57 but requires regulatory specification to ensure uniform implementation and audit clarity.

Proposed Actions to Operationalise Regulatory Flexibility
To formalise and enforce this initiative, the following actions will be implemented:

1. Introduction of a new regulatory clause permitting retrieval of unused medical aids under Section 57.
2. Issuance of a formal circular *(pekeliling)* to standardise nationwide implementation.
3. Legal review and compliance validation to ensure alignment with Act 4 and existing subsidiary regulations.
4. Formal asset reclassification framework distinguishing:
    - Permanently Issued Medical Aid
    - Conditionally Issued (Recoverable) Medical Aid

Expected Outputs

- Official regulatory directive ("black and white" authority).
- Approved governance framework supported by Standard Operating Procedures (SOPs).
- Clear asset lifecycle classification.
- Enforceable nationwide compliance structure.

With written authority in place, structured compliance across all Local Offices can reasonably be expected.

Suggested Draft Clauses (For Board Consideration)
Proposed Clause 1: Conditional Issuance

Medical aids provided under Section 57 shall remain under the ownership of the Organisation unless otherwise determined. Such medical aids may be issued on a conditional usage basis for the duration of medical necessity.

Proposed Clause 2: Right of Retrieval

The Organisation reserves the right to retrieve medical aids issued under Section 57 upon cessation of medical necessity, recovery, permanent transition to alternative aids, case closure, or death of the insured person.

Proposed Clause 3: Refurbishment and Reallocation

Retrieved medical aids may be refurbished, certified for safety and functionality, and reassigned to other eligible insured persons in accordance with approved standards and procedures.

Proposed Clause 4: Asset Classification

Medical aids shall be classified as either:

(a) Permanently Issued Medical Aid; or

(b) Conditionally Issued (Recoverable) Medical Aid,

as determined by the Organisation based on clinical and operational criteria.

These clauses are:

- Legally defensible under Section 57(2) ("on such conditions as may be specified")
- Audit-friendly
- ESG-aligned
- Operationally enforceable

4.1.2 Phase 2 (Month 3–8): Extend INSPIRE

It was considered most appropriate for PERKESO to introduce this new initiative through organisation-wide synchronisation. Therefore, it was determined that the extension of the

INSPIRE system should be initiated first as this would ensure that all INSPIRE users are formally informed and aligned with the new framework.

Current INSPIRE System
(Accessible only by Case Managers and PERKESO Doctors)

1. Medical aid orders for *Orang Berinsurans (OBs)*, such as prosthetics and wheelchairs, are submitted into the system by case managers at Local Offices based on hospital doctors' reports.
2. The requests are then assessed by PERKESO doctors from *BPPP (Bahagian Pencegahan, Perubatan, Pemulihan)*, and approvals are granted where appropriate.
3. Once approval has been obtained, the orders are processed by the case managers.

Workflow – Case Manager (Local Office)

1. Different categories of cases, including ILAT, HUS, and HUK, are received by Local Office case managers (CMs).
2. An initial assessment is conducted through review of diagnosis and medical history.
3. Workplace visits are carried out, and determinations are made as to whether the rehabilitation goal is Return-to-Work (RTW) or Activity of Daily Living (ADL).
4. Job modification, job capacity evaluation, and work ability assessments are performed.
5. Monitoring is conducted for approximately 1–3 months to assess sustainability of the OB at work.
6. Case management is typically maintained for up to one year until case closure.
7. If rehabilitation intervention is required, the OB is referred to the rehabilitation team by the local office case manager.

Workflow – Case Manager (Rehabilitation)

1. Focus is placed on the OB's physical healing progress for approximately 3–6 months.
2. Initial assessments are conducted, and rehabilitation plans are developed based on rehabilitation goals (working or non-working).
3. Treatment plans are implemented according to diagnosis.

4. Upon discharge, notification is provided to the Local Office CM.

Identified Monitoring Gap

At present, monitoring of medical aid usage is conducted primarily through the PHU *(Pemulihan Hilang Upaya)* Local Office and rehabilitation case managers. Monitoring mechanisms for *Orang Berinsurans (OBs)* who have fully recovered or who have passed away are not systematically integrated into the INSPIRE system.

For PHU Local Office case managers, OBs who participate in PHU programme are closely monitored. However, the usability of medical aids is not monitored for OBs who do not participate in the PHU programme as medical aids are issued without structured follow-up of healing progression, effectively resulting in a supply-and-forget model without lifecycle tracking.

According to Local Office Putrajaya, non-PHU cases constitute most cases managed. Since non-PHU cases are higher, it creates even more material risk of medical aid underutilisation, premature replacement and avoidable procurement expenditure due to the absence of a structured lifecycle monitoring framework.

Proposed INSPIRE Prototype Extension

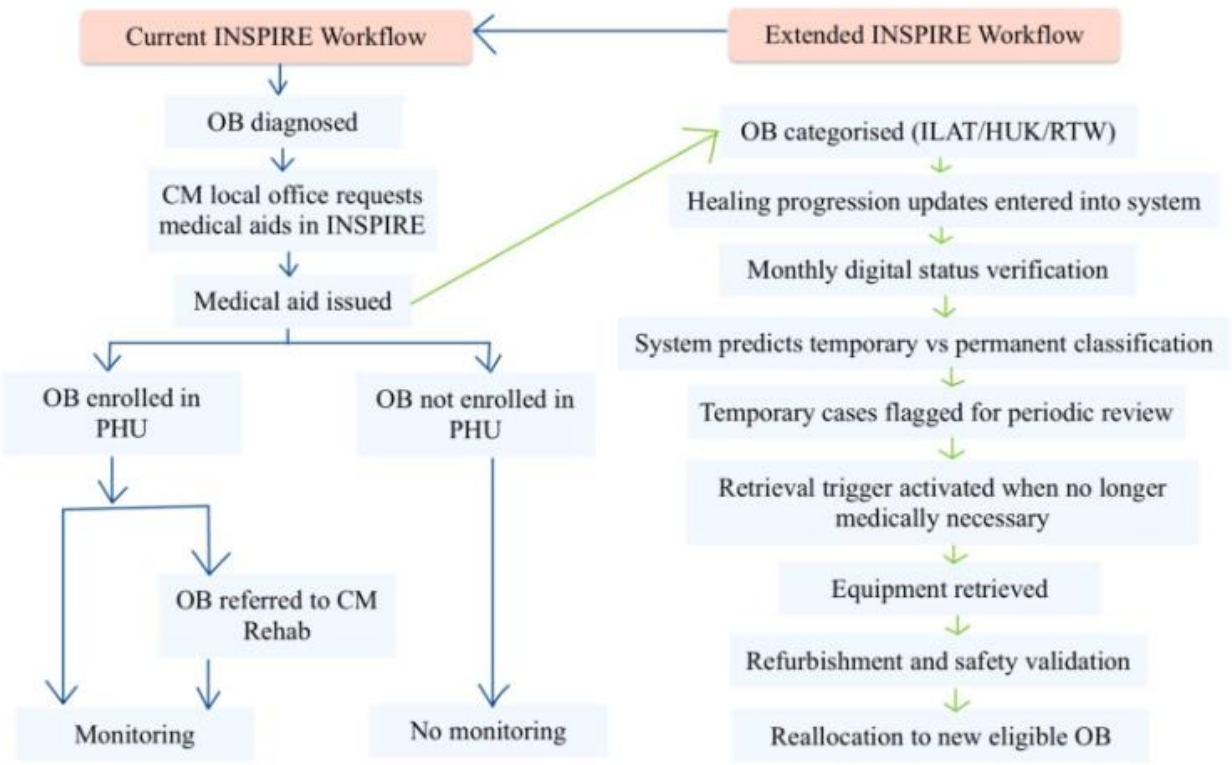

Figure 5: Extended INSPIRE Workflow

Under the proposed extension:

1. Data on OB progression will be incorporated into INSPIRE to enable early identification of medical aids that may be eligible for retrieval.
2. Updates on OB progression will be entered by case managers through a newly developed system function.
3. OBs will be categorised into structured groups such as ILAT, HUK, and RTW to allow systematic monitoring and predictive tracking.
4. Based on historical case data and predictive modelling, early determination will be made as to whether a medical aid is temporary or permanent in nature.
5. Temporary cases will be automatically flagged for periodic review while permanent cases may be subjected to reduced monitoring frequency, with structured verification intervals established where appropriate.

6. A monthly digital verification mechanism may be incorporated, whereby case status (active, recovered, deceased, or transitioned) is updated within the system.
7. Predictions regarding the expected duration of medical aid usage will be generated based on diagnosis patterns and historical trends.

For example, for a stroke patient, initial wheelchair usage may be required, followed by transition to a walking frame as mobility improves. The wheelchair may be retrieved prior to issuance of the walking frame, optimising resource allocation.

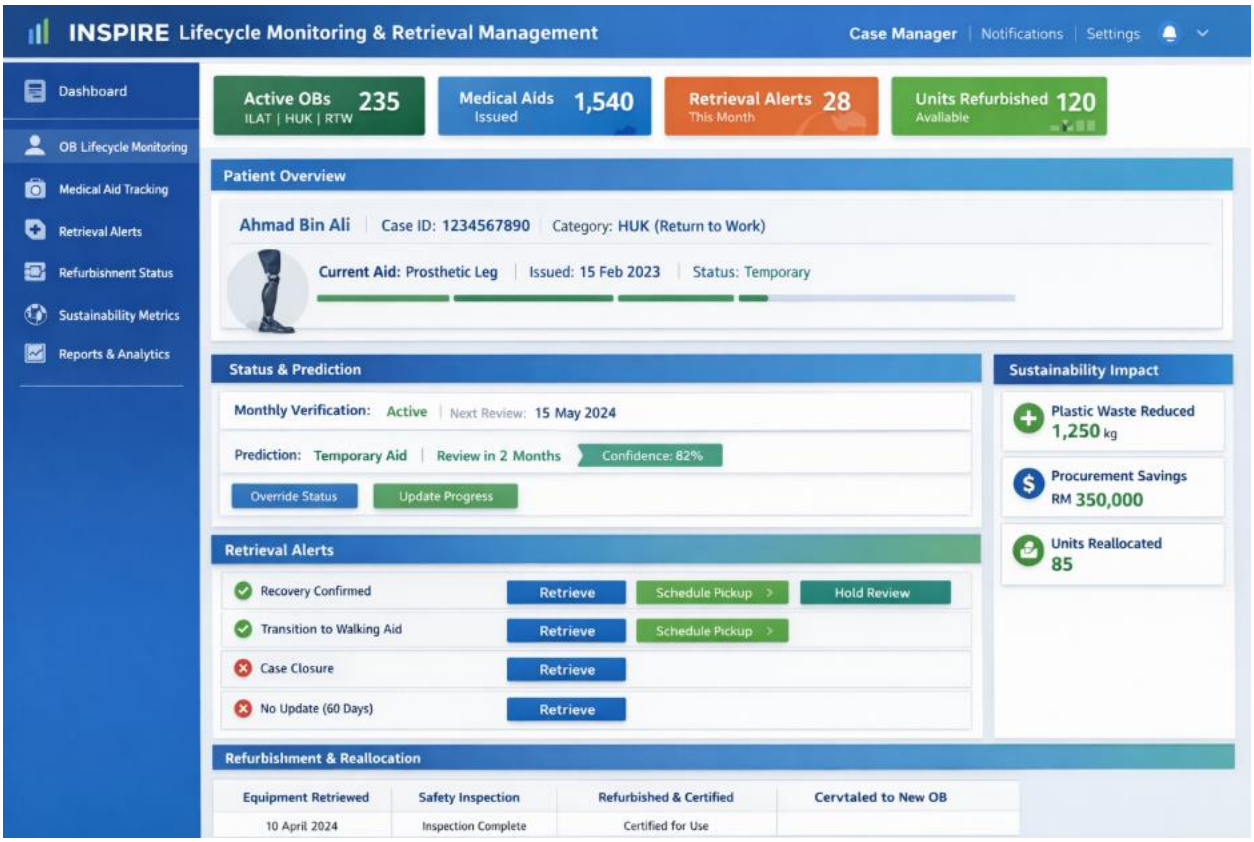

Figure 6: Proposed Extension of INSPIRE

4.1.3 Phase 3 (Month 8-14): Pilot Implementation, Refurbishment Workflow & Workforce Formation

Objective

To validate operational feasibility, establish refurbishment capability, and formally create the Refurbishment Team structure.

Problems Identified

During the rotation at *Pusat Rehabilitasi PERKESO Tun Abdul Razak* and Putrajaya Local Office, it was further noticed that:

- Storage at Pusat Rehabilitasi PERKESO Tun Abdul Razak is unstructured and manually managed.
- Many unrecorded delivery delays are reported at Putrajaya Local Office with one case still not receiving the medical aid since September 2025.
- No structured refurbishment workflow exists.
- No empirical data exists on retrieval efficiency.
- No dedicated team is responsible for lifecycle management of medical aids.

ADD GAMBAR STORAGE UNITS

Pilot implementation at:

- Putrajaya Local Office
- Pusat Rehabilitasi PERKESO Tun Abdul Razak

Refurbishment Team Creation (New Structured Unit)

A dedicated Medical Aid Refurbishment Unit will be formally established.

Team Composition:

1. Medical equipment technicians
2. Sterilisation officers
3. Safety compliance officers
4. Inventory control analysts
5. Logistics coordinators
6. Quality assurance personnel

Hygiene Assurance Framework

During informal engagements and feedback sessions with Orang Berinsurans (OBs), concerns were expressed regarding the hygiene and safety of refurbished medical equipment.

These concerns will be addressed through:

International Benchmark – New Zealand ACC Model

Under the Accident Compensation Corporation circular rehabilitation framework:

- Equipment is systematically retrieved.
- Advanced sterilisation technologies are applied.
- Equipment undergoes safety validation before redistribution.

Proposed Sterilisation Standards

Within PERKESO's Refurbishment Unit:

- Industrial-grade cleaning procedures
- Advanced UV sterilisation technology

- Disinfection chambers
- Component-level sanitisation
- Contamination risk assessment

Mandatory Testing & Certification Protocol

Before redistribution, equipment must undergo:

- Safety inspection
- Functional performance testing
- Mechanical integrity verification
- Load-bearing validation
- Electrical safety testing (if applicable)
- Quality assurance sign-off
- Certification tagging
- Digital re-registration in INSPIRE

No equipment shall be redistributed without formal clearance.

Workforce Impact

This phase will:

- Create new skilled roles within PERKESO.
- Develop internal technical capability.
- Reduce dependency on external procurement cycles.
- Strengthen institutional refurbishment expertise.

This aligns with national workforce development objectives.

4.1.4 PHASE 4 (Month 14–24)

National Storage Restructuring & AI Integration

Problem

Current system is:

- Decentralised
- Manually managed
- Non-standardised
- Not AI-integrated
- Without lifecycle monitoring

Resulting in delays, overstocking, wastage, and poor traceability.

International Benchmark Comparison

The proposed national storage and AI-enabled logistics system will be modelled after internationally recognised automated warehouse platforms, specifically those developed by Swisslog and Kardex International. These companies have long-standing track records in intralogistics automation, delivering measurable improvements in efficiency, accuracy, traceability, and sustainability across multiple industries, including healthcare supply chains.

Swisslog – AI-Driven Intralogistics and Automation

Swisslog is a global leader in automated warehouse and logistics solutions, offering modular, software-driven systems that integrate robotics, data analytics, and intelligent warehouse management into a cohesive platform. Its solutions are widely used to support real-time tracking, predictive demand forecasting, automated storage and retrieval, and operational optimisation across complex logistics networks.

Key strengths of Swisslog's systems include:

- **End-to-end automation and data integration:** Swisslog's SynQ software platform unifies inventory control, automation hardware, and operational data into a single control

layer. This enables precise real-time visibility and reduces manual errors that are common in manual warehouses.

- **Compliance and traceability improvements:** In regulated industries such as pharmaceuticals and healthcare, automation enhances compliance and traceability, ensuring that every item's movement and status can be audited with confidence.
- **Increased operational efficiency:** Automated systems reduce reliance on manual handling, increase throughput, and optimise material flow — leading to faster response times and lower operational risk.
- **Reduced operational risk and improved workforce safety:** By automating repetitive and potentially hazardous tasks, organisations reduce exposure to workforce shortages and injuries, while staff can be redeployed to higher-value activities such as quality assurance and data analysis.

Swisslog's extensive global case studies demonstrate that organisations using its automated warehouse solutions typically achieve:

- Higher storage density and improved space utilisation.
- Reduction in picking and retrieval time.
- Better compliance and traceability, especially critical where quality and safety are regulated.

These benefits are directly aligned with PERKESO's goals of lifecycle monitoring, retrieval flagging, and seamless refurbishment redistribution.

Kardex International – Precision Storage & Controlled Environments

Kardex International is a recognised manufacturer of automated storage and retrieval systems (ASRS) tailored for environments that require precise control, high availability, and efficient space utilisation. Its solutions have been widely adopted in hospital and medical supply contexts where:

- **Controlled storage conditions are essential:** Kardex systems can provide cleanroom-compatible environments with controlled temperature and humidity, which help preserve sensitive medical equipment and supplies.
- **Space efficiency dramatically increases:** Automated systems can reduce physical footprint by up to 85% compared with traditional shelving, enabling significant space optimisation in central warehouses or regional sub-stores.
- **Real-time inventory control:** Stock levels and item movements can be monitored continuously, with AI-enabled features alerting operators to low stock, retrieval eligibility, or expiring items.
- **Improved operational speed and accuracy:** Automatic vertical transport and retrieval reduce manual travel time and material handling mistakes, increasing reliability and throughput.

For example, Kardex's specialised hospital bed storage solutions not only maximise space but also ensure that critical healthcare supplies are stored in dust-free, condition-controlled environments suitable for sensitive medical items.

Multi-Tiered National Inventory Architecture

To address structural inefficiencies, a multi-tiered, AI-enabled inventory infrastructure will be implemented as seen on the diagrams above. The 3D diagrams visually illustrate this architecture, modelled in alignment with internationally recognised automated intralogistics platforms developed by Swisslog and Kardex International.

The structure below explains precisely how each tier follows those international benchmarks.

Tier 1 – Central AI Warehouse & Refurbishment Hub

Role

National distribution centre and central refurbishment hub.

This tier mirrors the **centralised automation control model** used in Swisslog's integrated warehouse ecosystems and the **high-density controlled storage model** deployed by Kardex in healthcare environments.

Core Functions

- Bulk storage of all medical aids
- National redistribution control
- Refurbishment operations
- Lifecycle governance oversight

Dedicated Refurbishment Zones

- Cleaning and UV sterilisation chambers
- Component repair benches
- Safety and functional quality inspection stations
- Certification tagging and documentation areas
- Quarantine holding zone
- Quality control command centre

International Benchmark Alignment

1. **Controlled Sterilisation & Environmental Segmentation**
   - Closely follows Kardex's cleanroom-compatible and dust-controlled storage principles used in medical supply environments.
2. **Quality Control Command Centre**
   - Equivalent to Swisslog's AI-driven warehouse control tower, enabling:

- Real-time monitoring
- Audit tracking
- Predictive lifecycle alerts

Technology Integration

- AI-enabled warehouse management system
- Swisslog-style SynQ integration logic
- Kardex-style vertical lift modules
- Automated Storage & Retrieval Systems (ASRS)
- Robotics-assisted picking
- Real-time item-level tracking

International Benchmark Alignment

1. **Vertical Lift Modules & ASRS**

- Directly inspired by Kardex's high-density vertical storage systems, which:
- Reduce physical footprint
- Increase storage density
- Optimise ceiling height utilisation

2. **Robotics-Assisted Picking**

- Follows Swisslog's automation philosophy of reducing manual handling and improving throughput.

Tier 2 – Regional Sub-Stores

Role

Buffer and response acceleration layer.

Functions

- Buffer stock locations

- Temporary holding of retrieved equipment (maintain traceability continuity before refurbishment)
- Redistribution staging

Features

- Standardised shelving systems
- QR/RFID scanning
- AI-linked stock visibility
- Climate-controlled storage (where required)

Tier 2 ensures:

- Faster regional turnaround
- Reduced transport lag
- Without losing central governance

Tier 3 – Points of Use (Local Offices / Rehab Centres)

Role

Immediate clinical access layer. This tier represents the demand-facing layer of the system.

Functions

- Immediate access for case managers and clinicians
- Emergency allocation capability
- Direct INSPIRE integration

4.2 LONG-TERM PLAN (Month 24-36)

4.2.1 PHASE 5 (Month 24–36): Digital Marketplace and National Rollout

A Medical Aid Marketplace module will be integrated within Rehab Nav. It represents the demand–supply digital interface layer of the circular rehabilitation infrastructure.

Marketplace Features

- Listing of available refurbished equipment by local office
- Listing of refurbished equipment by location
- Proximity-based matching
- Real-time availability display
- Digital request and approval system
- Delivery options
- Self-collection options
- Functional Design Logic

Listing of refurbished equipment by local office / location

Ensures transparent visibility of available medical aids across the national network.

Proximity-based matching

Matches demand requests with the nearest available refurbished unit to reduce transport lag and optimise redistribution efficiency.

Digital request and approval system

Maintains governance control, enabling structured approvals while preserving compliance and audit trails.

Real-time availability display

Synchronised with Tier 1 (Central AI Warehouse) and Tier 2 (Regional Sub-Stores), ensuring up-to-date inventory visibility.

Delivery or self-collection options

Introduces operational flexibility while maintaining tracking continuity.